Autonomous Indoor Mobile Robot using ROS2 and Gazebo

Development and Implementation for Indoor Navigation

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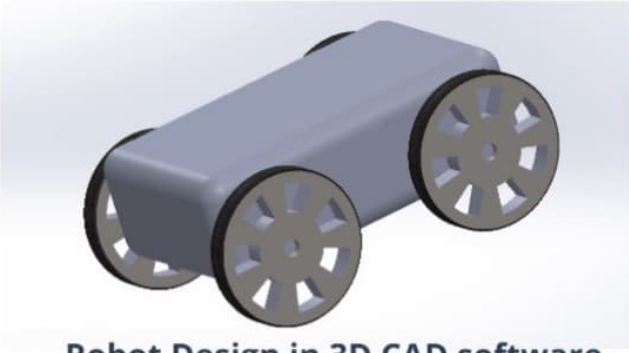
Project Overview

Problem Statement:
Develop a fully autonomous indoor robot capable of mapping and navigation in unknown environments.



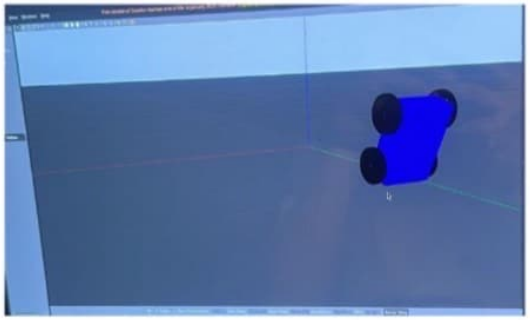
Robot Design in 3D CAD software

Objective :

- Design and simulate a mobile robot for indoor navigation
- Enable both manual control and autonomous behavior

Key Features:

- Teleoperation using keyboard (/cmd_vel)
- PID-based motion control
- SLAM for mapping and localization
- Obstacle detection using LIDAR



Robot Imported into Gazebo emty world

Technologies Used:
ROS2 Humble (middleware)
Gazebo Classic (simulation)
URDF (robot modeling)
RViz2 (visualization)
Nav2 (autonomous navigation)
SLAM Toolbox (mapping)

Team Contribution :
Robot Modeling & URDF : Shiwang and Saam
Simulation & World Design : Shiwang and Pranjal
Control (Teleop + PID) : Pranjal
SLAM & Mapping : Saam and Pranjal

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Robot Design and Specifications

Design Justification

- Skid-steer chosen for simplicity and robustness
- 4-wheel base provides stability during turns
- LiDAR selected for accurate indoor mapping

Robot Design

- **Drive Type** : Skid-steer (4-wheel differential drive)
- **Structure** : Base + 4 wheels + LIDAR sensor
- **Modeling Tool** : URDF (exported from SolidWorks)

Physical Parameters

- Base Dimensions: 0.30 × 0.20 × 0.08 meters (L×W×H)
- Mass: Approximately 2.5–3 kilograms
- Wheels Radius : 0.1335 meters, Wheel Separation : 0.20 – 0.30 m

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System Architecture & Data Flow

Architecture

Simulation environment:
Gazebo (robot + apartment world)

Robot Model:
URDF (links, joints, sensors)

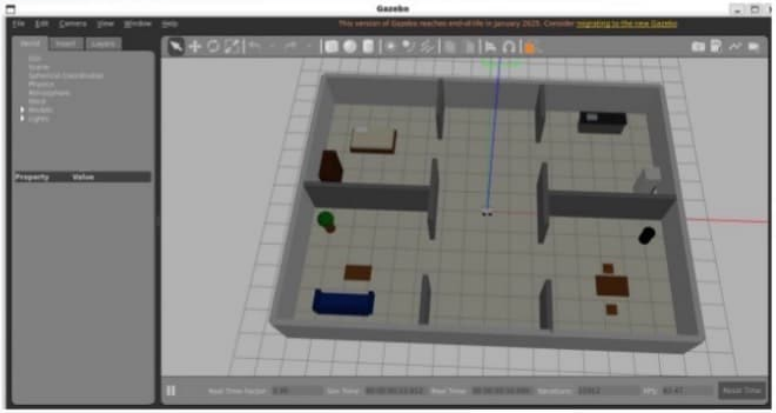
Middleware:
ROS2 communication

Control Layer:
Teleop + PID controller

Perception Layer:
LIDAR + SLAM

ROS2 Communication (Topics)

- **/cmd_vel** : velocity commands (control input)
- **/odom** : robot position & feedback
- **/scan** : LIDAR sensor data
- **/tf, /tf_static** : coordinate transformations



apartment.world

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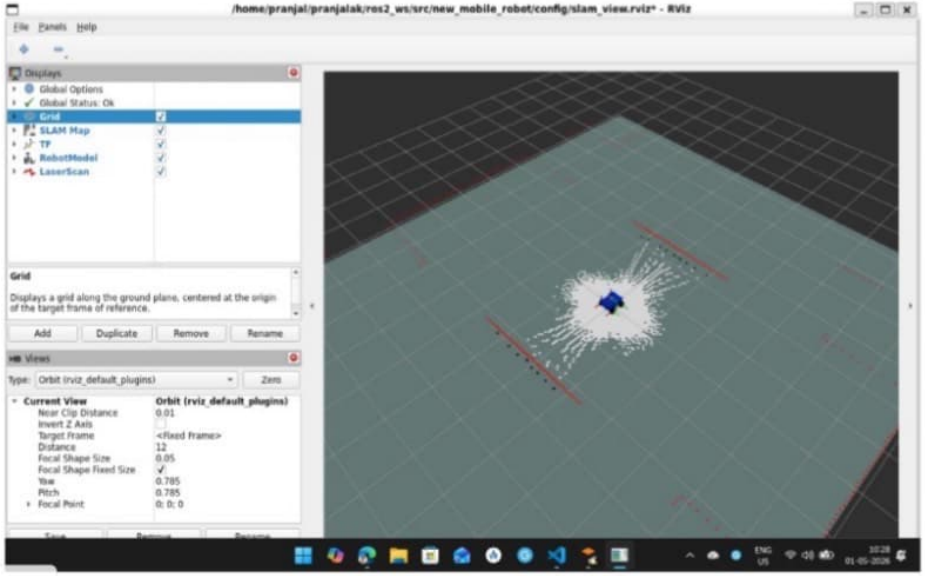
Results & Conclusion



Robot slam tracking

Results Achieved

- Successful robot simulation in Gazebo
- Smooth teleoperation via /cmd_vel
- Robot navigates between rooms and avoids obstacles

Performance Highlights

- Low control latency (< 50 ms)
- Stable trajectory (minimal oscillations)
- Real-time map generation in RViz2
- Reliable ROS2 communication between modules

Conclusion

- Complete indoor mobile robot system successfully implemented
- Demonstrates full robotics pipeline: Modeling --> Simulation --> Control --> Mapping
- System is scalable for real-world robotic applications

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Closing & Credits

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